ORDINARY DIFFERENTIAL EQUATIONS • UNIT 3.01

C Second-Order Linear Homogeneous ODEs & Wronskian

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

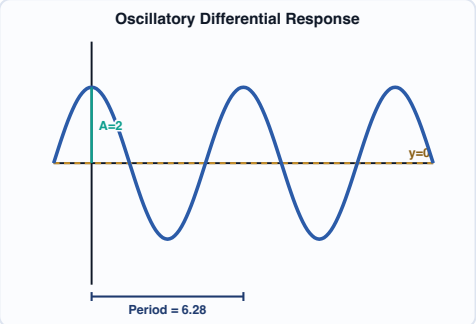
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 D	Q3 D	Q4 A	Q5 A
Q6 A	Q7 D	Q8 A	Q9 C	Q10 C
Q11 B	Q12 D	Q13 D	Q14 B	Q15 A
Q16 D	Q17 A	Q18 B	Q19 B	Q20 D
Q21 D	Q22 D	Q23 A	Q24 D	Q25 D

Second-Order Linear Homogeneous ODEs & Wronskian Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



- **A** 39 ✓
- **B** 24
- **C** 42
- **D** 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 39\lambda + 350 = 0$, the sum of roots is 39.

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

B $e^2 - 1$

C $3 - 2e^{-2}$

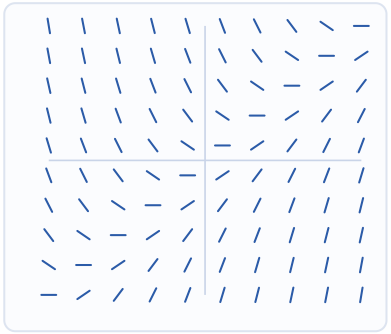
D $3 - 2e^{-1}$ ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$.
Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$.
Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$.
The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$.
Integrating both sides with respect to x: $ye^{2x} = \int 6e^{2x}dx$.
$ye^{2x} = 6 \cdot \frac{1}{2}e^{2x} + C$.
$ye^{2x} = 3e^{2x} + C$.
Solving for y: $y = 3 + Ce^{-2x}$.
Using the initial condition $y(0) = 1$ to find C: $1 = 3 + Ce^{-2(0)}$.
$1 = 3 + C$.
$C = 1 - 3 = -2$.
So, the particular solution is $y(x) = 3 - 2e^{-2x}$.
Now, we need to find $y(1)$: $y(1) = 3 - 2e^{-2(1)} = 3 - 2e^{-2}$.
This matches choice C.

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D **1.220** ✓

WORKED SOLUTION & VERIFICATION

Using Euler's method $y_{n+1} = y_n + h(x_n + y_n)$ with $x_0 = 0, y_0 = 1, h = 0.1$:
1. $y_1 = 1 + 0.1(0 + 1) = 1.100$ at $x_1 = 0.1$.
2. $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.220$ at $x_2 = 0.2$.

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

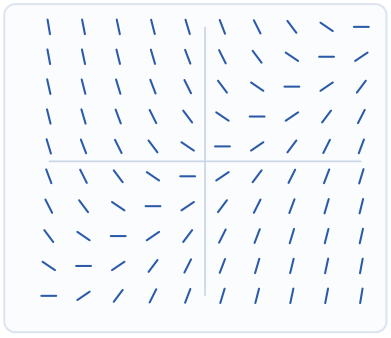
- **A** 5 ✓
- **B** -5
- **C** 6
- **D** -6

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation $r^2 - 5r + 6 = 0$, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $r_1 + r_2 = -\frac{-5}{1} = 5$.

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



- **A** 7 ✓
- **B** 6
- **C** 8
- **D** 5

WORKED SOLUTION & VERIFICATION

The differential equation is $\frac{dy}{dx} = 2x$. This is a separable equation. We can rewrite it as $dy = 2x\,dx$. Integrating both sides gives $\int dy = \int 2x\,dx$, which results in $y = x^2 + C$, where C is the constant of integration. We are given the initial condition $y(0) = 3$. Substituting these values into the general solution, we get $3 = (0)^2 + C$, so $C = 3$. Therefore, the particular solution is $y = x^2 + 3$. We need to find the value of $y(2)$. Substituting $x = 2$ into the particular solution, we have $y(2) = (2)^2 + 3 = 4 + 3 = 7$.

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

- **A** $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓
- **B** $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$
- **C** $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$
- **D** $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of x to be the same. Let $k = n-2$ in the first series, so $n = k+2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n+1$ in the second series, so $n = k-1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all x, the coefficients of each power of x must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing k with n, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** 24
- **D** 40 ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is 40.

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

- **A** 170 ✓
- **B** 151
- **C** 173
- **D** 187

WORKED SOLUTION & VERIFICATION

Substituting $s = 20$ gives $\frac{170}{20-19} = 170$.

9 MCQ

The second-order linear homogeneous ODE $F''(w) - 50F'(w) + 621F(w) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 31

B 73

C 50 ✓

D 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 50\lambda + 621 = 0$, the sum of roots is 50.

10 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$.
If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

A $k = -5$

B $k = 1$

C $k = 5$ ✓

D $k = -1$

WORKED SOLUTION & VERIFICATION

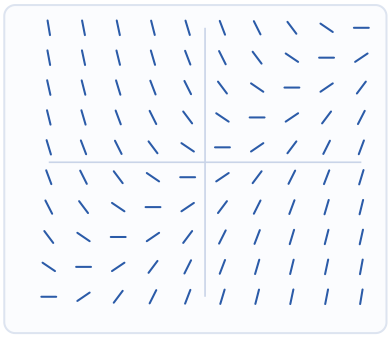
The given differential equation is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, the roots are repeated: $r_1 = r_2 = 1$.
One fundamental solution is given as $y_1(x) = e^x$. For a second-order linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$ with a known solution $y_1(x)$, the Wronskian is given by $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x) = Ce^{-\int p(x)dx}$.
In our equation, $p(x) = -2$. So, $-\int p(x)dx = -\int(-2)dx = 2x$. Therefore, $W(y_1, y_2)(x) = Ce^{2x}$.
We are given that $W(y_1, y_2)(x) = 5e^{2x}$, which means $C = 5$.
We are also given $y_1(x) = e^x$ and $y_2(x) = kxe^x$.
Let's compute the Wronskian using these forms:
$y_1'(x) = e^x$
$y_2'(x) = \frac{d}{dx}(kxe^x) = k(1 \cdot e^x + x \cdot e^x) = ke^x(1+x)$
Now, substitute into the Wronskian formula:
$W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$
$W(y_1, y_2)(x) = e^x \cdot [ke^x(1+x)] - e^x \cdot [kxe^x]$
$W(y_1, y_2)(x) = ke^{2x}(1+x) - kxe^{2x}$
$W(y_1, y_2)(x) = ke^{2x} + kxe^{2x} - kxe^{2x}$
$W(y_1, y_2)(x) = ke^{2x}$
We equate this to the given Wronskian:
$ke^{2x} = 5e^{2x}$
Therefore, $k = 5$.

11 MCQ

Consider the second-order linear homogeneous differential equation

$$y'' - 3y' + 2y = 0.$$

Find the general solution to this equation.



A $y(x) = c_1 e^{-x} + c_2 e^{-2x}$

B $y(x) = c_1 e^{x} + c_2 e^{2x}$ ✓

C $y(x) = c_1 \cos(x) + c_2 \sin(x)$

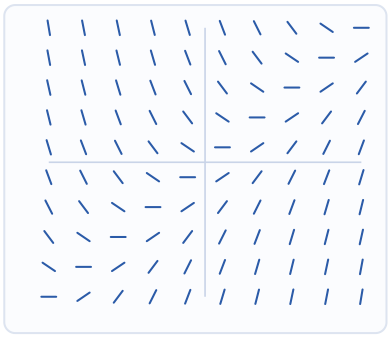
D $y(x) = c_1 x e^{x} + c_2 e^{x}$

WORKED SOLUTION & VERIFICATION

The characteristic equation for the differential equation $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = 0$. Factoring this quadratic equation gives $(r-1)(r-2) = 0$. The roots are $r_1 = 1$ and $r_2 = 2$. Since the roots are real and distinct, the general solution is of the form $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$. Substituting the roots, we get $y(x) = c_1 e^{x} + c_2 e^{2x}$.
Choice B is incorrect because it uses negative roots.
Choice C is incorrect because it corresponds to complex roots.
Choice D is incorrect because it corresponds to a repeated root.

12 MCQ

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.



A $A = \frac{1}{2}, B = 0$

B $A = 0, B = \frac{1}{2}$

C $A = -\frac{1}{2}, B = 0$

D $A = 0, B = -\frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - y = \sin x$ and a proposed particular solution $y_p(x) = A\cos x + B\sin x$.

We need to find the derivatives of $y_p(x)$ and substitute them into the differential equation.

First derivative: $y_p'(x) = -A\sin x + B\cos x$.

Second derivative: $y_p''(x) = -A\cos x - B\sin x$.

Substitute these into the differential equation:

$(-A\cos x - B\sin x) - (A\cos x + B\sin x) = \sin x$

Combine like terms:

$(-A - A)\cos x + (-B - B)\sin x = \sin x$

$-2A\cos x - 2B\sin x = \sin x$

For this equation to hold for all x, the coefficients of $\cos x$ and $\sin x$ on both sides must be equal.

Comparing coefficients of $\cos x$: $-2A = 0$, which implies $A = 0$.

Comparing coefficients of $\sin x$: $-2B = 1$, which implies $B = -\frac{1}{2}$.

Therefore, the values of the constants are $A = 0$ and $B = -\frac{1}{2}$.

13 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation.

(a) Find the value of the parameter k.

(b) Find the general solution to the differential equation.

A $k = 3, y(x) = c_1e^{3x} + c_2xe^{3x}$

B $k = -3, y(x) = c_1e^{3x} + c_2e^{x}$

C $k = 9, y(x) = c_1e^{3x} + c_2e^{-3x}$

D $k = -3, y(x) = c_1e^{3x} + c_2e^{-x}$ ✓

WORKED SOLUTION & VERIFICATION

Part (a):

If $y_1(x) = e^{3x}$ is a solution, then its derivatives must satisfy the equation.

$y_1'(x) = 3e^{3x}$

$y_1''(x) = 9e^{3x}$

Substitute these into the differential equation $y'' - 2y' + ky = 0$:

$9e^{3x} - 2(3e^{3x}) + k(e^{3x}) = 0$

$9e^{3x} - 6e^{3x} + ke^{3x} = 0$

Factor out e^{3x} (which is never zero):

$e^{3x}(9 - 6 + k) = 0$

$3 + k = 0$

So, $k = -3$.

Part (b):

Now that we know $k = -3$, the differential equation is $y'' - 2y' - 3y = 0$.

The auxiliary (characteristic) equation is $r^2 - 2r - 3 = 0$.

Factor the quadratic equation: $(r - 3)(r + 1) = 0$.

The roots are $r_1 = 3$ and $r_2 = -1$.

Since the roots are real and distinct, the general solution is of the form $y(x) = c_1e^{r_1x} + c_2e^{r_2x}$.

Therefore, the general solution is $y(x) = c_1e^{3x} + c_2e^{-x}$.

This matches choice D.

14 MCQ

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

A $\frac{1}{4}$

B $\frac{1}{2}$ ✓

C $\frac{1}{2}\sin(2x)$

D $\frac{1}{4}\cos(2x)$

WORKED SOLUTION & VERIFICATION

The given differential equation is $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. The associated homogeneous equation is $\frac{d^2y}{dx^2} + 4y = 0$. The characteristic equation is $r^2 + 4 = 0$, which has roots $r = \pm 2i$. Thus, the fundamental set of solutions for the homogeneous equation is $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$.

We use the method of variation of parameters to find a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$.

The Wronskian of y_1 and y_2 is:

$W(y_1, y_2)(x) = y_1y_2' - y_1'y_2 = \cos(2x)(2\cos(2x)) - (-\sin(2x))(2\sin(2x))$

$W(y_1, y_2)(x) = 2\cos^2(2x) + 2\sin^2(2x) = 2(\cos^2(2x) + \sin^2(2x)) = 2.$

The non-homogeneous term is $g(x) = \sec(2x)$.

The formulas for $u_1'(x)$ and $u_2'(x)$ are:

$u_1'(x) = -\frac{y_2(x)g(x)}{W(x)}$

$u_2'(x) = \frac{y_1(x)g(x)}{W(x)}$

Substituting the given functions:

$u_2'(x) = \frac{\cos(2x)\cdot\sec(2x)}{2}$

$u_2'(x) = \frac{\cos(2x)}{2\cos(2x)}$

$u_2'(x) = \frac{1}{2}.$

The value of $u_2'(x)$ is $\frac{1}{2}$.

15 MCQ

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?

A -5 ✓

B -6

C 5

D 6

WORKED SOLUTION & VERIFICATION

The general form of a second-order linear homogeneous differential equation with constant coefficients is $ay''(x) + by'(x) + cy(x) = 0$. The associated auxiliary equation is $ar^2 + br + c = 0$. For the given equation $y''(x) + 5y'(x) + 6y(x) = 0$, the auxiliary equation is $r^2 + 5r + 6 = 0$. According to Vieta's formulas, for a quadratic equation of the form $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2 = -b/a$. In this case, $a = 1$, $b = 5$, and $c = 6$. Therefore, the sum of the roots is $r_1 + r_2 = -5/1 = -5$.

16 MCQ

The separable initial value problem $\frac{dp}{dt} = 34t$ satisfies $p(0) = 5$. Determine the solution value $p(18)$.

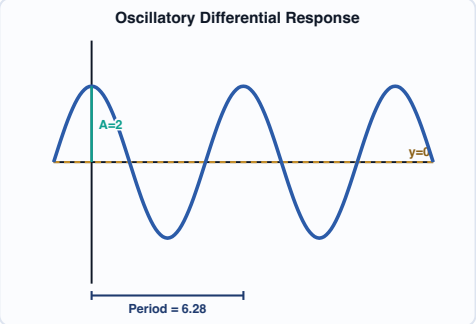
- A 5530
- B 5508
- C 5516
- D **5513** ✓

WORKED SOLUTION & VERIFICATION

Integrating gives $p(t) = 17t^2 + 5$, so $p(18) = 5513$.

17 MCQ

The second-order linear homogeneous ODE $p''(t) - 41p'(t) + 210p(t) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



- A **41** ✓
- B 21
- C 47
- D 44

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 41\lambda + 210 = 0$, the sum of roots is 41.

18 MCQ

The Laplace transform of $p(t) = 6e^{17t}$ is $q(s) = \frac{6}{s - 17}$ for $s > 17$. Determine the evaluated transform $q(18)$.

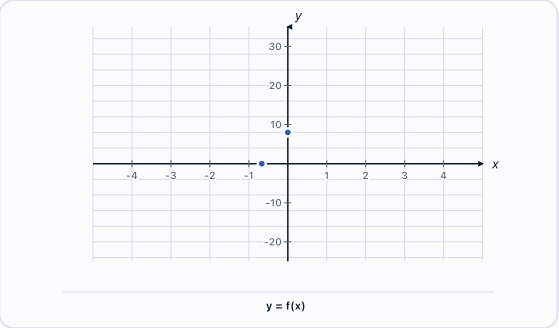
- A -11
- B **6** ✓
- C 9
- D 8

WORKED SOLUTION & VERIFICATION

Substituting $s = 18$ gives $\frac{6}{18-17} = 6$.

19 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.



A $A = 3$

B $A = 2$ ✓

C $A = 6$

D $A = 12$

WORKED SOLUTION & VERIFICATION

The given nonhomogeneous differential equation is $y'' - 5y' + 6y = 12x + 8$. \nThe form of the nonhomogeneous term is $g(x) = 12x + 8$, which is a polynomial of degree 1. Thus, we assume a particular solution of the form $y_p(x) = Ax + B$.

We need to find the first and second derivatives of $y_p(x)$:

$y_p'(x) = A$

$y_p''(x) = 0$

Substitute these into the differential equation:

$0 - 5(A) + 6(Ax + B) = 12x + 8$

$-5A + 6Ax + 6B = 12x + 8$

$6Ax + (6B - 5A) = 12x + 8$

Equating the coefficients of like powers of x:

For the x term: $6A = 12 \Rightarrow A = 2$.

For the constant term: $6B - 5A = 8$.

Substitute $A = 2$ into the second equation:

$6B - 5(2) = 8$

$6B - 10 = 8$

$6B = 18$

$B = 3$

Therefore, the particular solution is $y_p(x) = 2x + 3$. The value of the constant A is 2.

20 MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 45

B 50

C 23

D **42** ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 42\lambda + 272 = 0$, the sum of roots is 42.

21 MCQ

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, where ν is a constant. This is Bessel's equation of order ν. Assume $\nu = 1/2$ and that $x > 0$.

Part 1: Ordinary Point Analysis

$x = 0$ is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

A Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

B Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

C Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

D **Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$** ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is $x^2y'' + xy' + (x^2 - \nu^2)y = 0$. We can rewrite this in the standard form $y'' + P(x)y' + Q(x)y = 0$ by dividing by x^2: $y'' + \frac{1}{x}y' + \frac{x^2-\nu^2}{x^2}y = 0$. Thus, $P(x) = \frac{1}{x}$ and $Q(x) = \frac{x^2-\nu^2}{x^2} = 1 - \frac{\nu^2}{x^2}$. $x = 0$ is a singular point. To determine if it's a regular singular point, we examine $xP(x)$ and $x^2Q(x)$. $xP(x) = x \cdot \frac{1}{x} = 1$, which is analytic at $x = 0$. $x^2Q(x) = x^2 \cdot \frac{x^2-\nu^2}{x^2} = x^2 - \nu^2$, which is also analytic at $x = 0$. Therefore, $x = 0$ is a regular singular point. The method of Frobenius assumes a solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^{n+r}$, where $c_0 \neq 0$. Substituting this into the differential equation, we get the indicial equation, which is a quadratic equation in r. For Bessel's equation, the indicial equation is given by $r(r-1) + p_0 r + q_0 = 0$, where $p_0 = \lim_{x\to 0} xP(x)$ and $q_0 = \lim_{x\to 0} x^2Q(x)$. In our case, $p_0 = 1$ and $q_0 = -\nu^2$. So the indicial equation is $r(r-1) + 1 \cdot r - \nu^2 = 0$, which simplifies to $r^2 - r + r - \nu^2 = 0$, or $r^2 - \nu^2 = 0$. Given $\nu = 1/2$, the indicial equation becomes $r^2 - (1/2)^2 = 0$, which is $r^2 - \frac{1}{4} = 0$. The roots of this equation are $r = \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$.

22 MCQ

The second-order linear homogeneous ODE $u''(s) - 43u'(s) + 460u(s) = 0$ has characteristic roots λ_1, λ_2. Compute the trace sum $\lambda_1 + \lambda_2$.

A 32

B 46

C 66

D **43** ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 43\lambda + 460 = 0$, the sum of roots is 43.

23 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

A **2** ✓

B $2x$

C $2\sin(2x)$

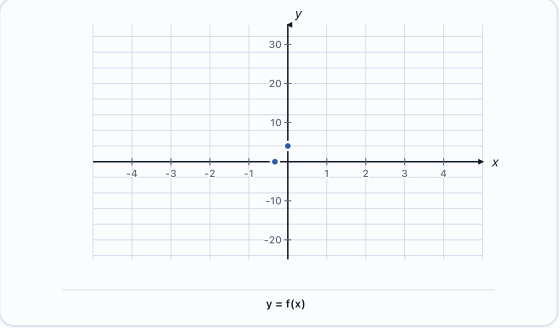
D $2\cos(2x)$

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' + 4y = 8$. This is a nonhomogeneous second-order linear differential equation. We use the method of undetermined coefficients to find a particular solution, $y_p(x)$. The nonhomogeneous term is a constant, 8. We guess a particular solution of the form $y_p(x) = A$, where A is a constant. Then, $y_p'(x) = 0$ and $y_p''(x) = 0$. Substituting these into the differential equation: $0 + 4(A) = 8$. Solving for A, we get $4A = 8$, which means $A = 2$. Therefore, the particular solution is $y_p(x) = 2$.

24 MCQ

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.



A 1

B 6

C 12

D **2** ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is $y'' - 5y' + 6y = 12x + 4$.
We are given that a particular solution is of the form $y_p(x) = Ax + B$.
To find the coefficients A and B, we substitute $y_p(x)$ and its derivatives into the differential equation.
The first derivative is $y_p'(x) = A$.
The second derivative is $y_p''(x) = 0$.
Substituting these into the differential equation:
$0 - 5(A) + 6(Ax + B) = 12x + 4$
$-5A + 6Ax + 6B = 12x + 4$
$6Ax + (6B - 5A) = 12x + 4$
By comparing the coefficients of x on both sides, we get:
$6A = 12 \implies A = 2$
By comparing the constant terms on both sides, we get:
$6B - 5A = 4$
$6B - 5(2) = 4$
$6B - 10 = 4$
$6B = 14 \implies B = \frac{14}{6} = \frac{7}{3}$
The question asks for the value of A. Therefore, $A = 2$.

25 MCQ

The second-order linear homogeneous ODE $M''(z) - 49M'(z) + 510M(z) = 0$ has characteristic roots λ_1, λ_2. Compute the trace sum $\lambda_1 + \lambda_2$.

A 64

B 52

C 33

D **49** ✓

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 49\lambda + 510 = 0$, the sum of roots is 49.